

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 2 – Technical Presentation

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO2	Assessment	Assessment Tool 2: Technical Presentation
Weightage	30% of CIA	Total Marks	100
CO1 Statement	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2,BL3).		
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Section/Batch	Slot C / 2025	Date	06-10-2026

Code of Conduct:

I, D Mahaboob Basha(192525369), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	



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TECHNICAL PRESENTATION

Stack ADT-Based Backward Navigation in Web Browsers

CSA0314-DATA STRUCTURES

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INTRODUCTION

- Web browsers allow users to visit multiple web pages.
- Backward navigation helps users return to previously visited pages.
- Stack ADT(Abstract Data Type) can be used to explain browser history management
- Google Chrome is a real-world example

Problem Statement

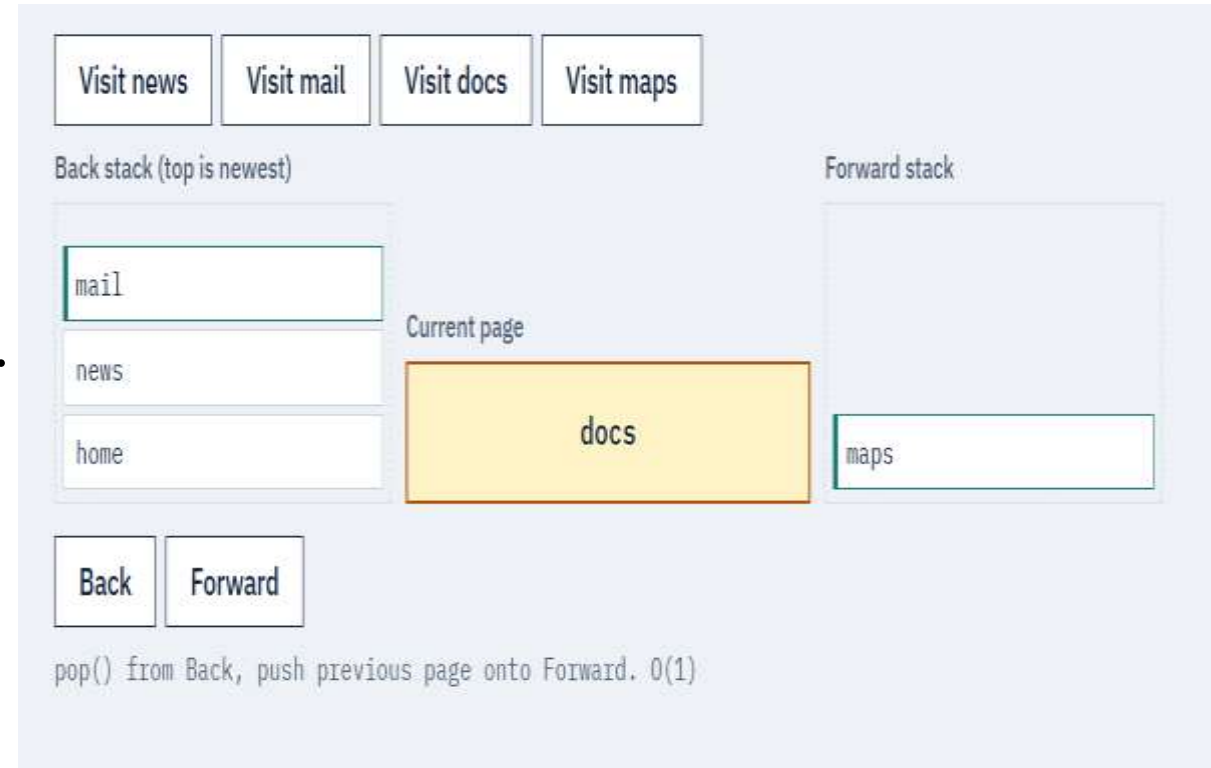
- Users visit multiple web pages in sequence.
- They may need to return to a previously visited page.
- Searching the entire browsing history is unnecessary for the immediate previous page.
- An efficient method is needed to manage navigation history.

Operations in Stack ADT

- Stack is a linear data structure.
- It follows LIFO (Last In, First Out).
- The last inserted element is removed first.

Basic operations:

- Push: Add an element.
- Pop: Remove the top element.
- Peek: View the top element.
- isEmpty: Check whether the stack is empty.



Performance Analysis of Stack

Operation	Time Complexity
Push	$O(1)$
Pop	$O(1)$
Peek	$O(1)$
Display	$O(n)$
Space Complexity	$O(n)$

$O(1)$: Constant time; the operation does not depend on the number of stored pages.

$O(n)$: Linear time; the time or space grows proportionally with the number of pages.

Applications

- Web browsers.
- Undo and redo operations in text editors.
- Mobile application navigation
- Function calls in programming

Conclusion

- Stack ADT is a suitable model for understanding backward navigation in web browsers.
- It follows the LIFO principle, making it easy to retrieve recently visited pages first.
- Stack operations like Push and Pop are efficient, typically taking $O(1)$ time.
- Stacks help organize data and are widely used in browsers, programming, and other applications.
- Understanding Stack ADT shows how simple data structures improve performance, organization, and user experience.

THANK YOU